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Jet flames of a refuse derived fuel

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ABSTRACT

This paper is concerned with combustion of a refuse derived fuel in a small-scale flame. The objective is to provide a direct comparison of the RDF flame properties with properties of pulverized coal flames fired under similar boundary conditions. Measurements of temperature, gas composition (O_2, CO_2, CO, NO) and burnout have demonstrated fundamental differences between the coal flames and the RDF flames. The pulverized coals ignite in the close vicinity of the burner and most of the combustion is completed within the first 300 ms. Despite the high volatile content of the RDF, its combustion extends far into the furnace and after 1.8 s residence time only a 94% burnout has been achieved. This effect has been attributed not only to the larger particle size of fluffy RDF particles but also to differences in RDF volatiles if compared to coal volatiles. Substantial amounts of oily tars have been observed in the RDF flames even though the flame temperatures exceeded 1300 °C. The presence of these tars has enhanced the slagging propensity of RDF flames and rapidly growing deposits of high carbon content have been observed.

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1. Introduction

Fossil fuels are still the main source of energy conversion for the power industry and they are likely to be dominant for decades to come. Co-firing of fuels which are regarded as neutral for emissions of carbon dioxide may be one of a few low-cost options to achieve reductions in carbon dioxide emissions. The currently occurring increase of the fuel and energy prices has resulted in increased interest in secondary/alternative fuels to the extent that has never been observed before. Both co-firing and complete fuel substitution belong to the most relevant issues in industrial combustion. In Germany sewage-sludge, saw-dust, agriculture wastes and in particular refuse derived fuels (RDF) have been combusted with coals in cement-making plants on a regular basis. In the power industry the substitution levels are restricted at present to about 5% or less.

In Germany, the new legislative regulations, introduced in 2005, allow for waste disposal through landfill for biologically neutral wastes only. Thus the municipal solid wastes, commercial and bulky wastes have to be processed either in mechanical–biological or mechanical–physical waste treatment plants. Refuse derived fuels produced as pellets or fluff are the principal products of these waste treatment processes. Co-firing of RDFs with fossil fuels has become not only an industrial necessity but it also opens up options for decreasing overall fuel costs since plant/boiler operators are remunerated for the RDF usage. In short, RDFs have become very important secondary fuels of Europe.

There exist several reviews on co-firing of coal with biomass fuel blends as exemplified by Refs. [1,2]. However there are perhaps only a few publications directly related to the burner design. Abbas et al. [3] used a 0.5 MW thermal input down-fired furnace to examine the influence of the fuel injection mode on coal–biomass flames. Pine saw-dust and dry pulverized sewage-sludge were combusted with bituminous coal. Both secondary fuels were supplied to the burner through separate channels and two burner configurations were examined. In the first configuration, named as annular primary fuel (APF), a central secondary fuel jet was surrounded by an annular orifice that supplied the pulverized coal. In the second configuration named as central primary fuel (CPF), the central jet of pulverized coal was surrounded by an annular biomass stream. Gerhardt et al. [4] also studied the combined combustion of municipal sewage-sludge in coal dust furnace with emphasis on fuel handling, combustion characteristics and NOx emissions. In their trials the sludge provided up to 80% of the 500 kW total thermal input. Van de Kamp and Morgan [5] carried out trials at 2 MW thermal input and burned straw, waste paper and municipal sewage-sludge with two bituminous coals. The co-firing ratio varied from zero to 100% for both the straw and the sludge. The paper [5] focused on establishing the effect of co-firing on NOx, SOx emissions and char burnout in Type I and Type II (IFRF classification [6]) flames.

We are aware of only two publications where RDF was co-fired with pulverized coal. Wolski et al. [7] examined the effect of RDF share on partitioning of trace metals while Kupka et al. [8] investigated slagging propensity of a 5% RDF/95% coal blend. Some preliminary experiments on RDF–coal firing in a cyclone combustor were also carried out within the scope of the EEC PowerFlame2

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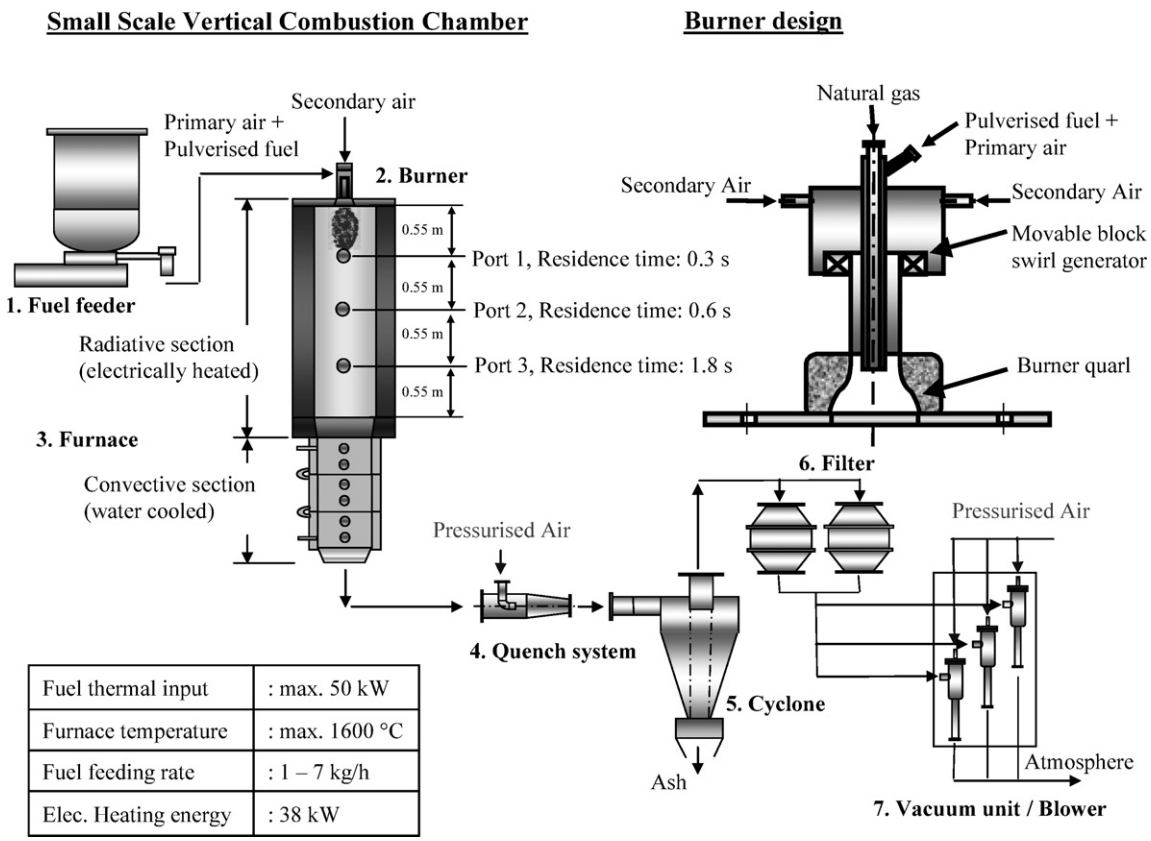

Fig. 1. A schematic diagram of the experimental rig.

project [9]. There exist several publications on chemical characterization of ashes produced during co-firing RDFs with coals as exemplified by the work of Norton et al. [10]. However, the literature on RDF firing is indeed scarce.

2. Objectives

Our paper is concerned with combustion of a refuse derived fuel in a small-scale flame. To the best of authors' knowledge this is the first time a RDF flame is created to establish its basic characteristics. Furthermore, we wish to provide a direct comparison of the RDF flame properties with properties of pulverized coal flames fired under similar boundary conditions. Such an approach facilitates a RDF characterization technique applicable to co-firing of pulverized fuels. There is an urgent demand for such a technique since the conventional fuel characterization methods based on TGA [11,12] are of a limited applicability to pulverized fuel firing. Similarly, the RDF combustion characterization in fixed beds [13–15] although informative and useful for grate fired stockers, is practically of no help for pulverized fuel firing.

3. Experimental

The essential parts of the rig (see Fig. 1) are: the fuel feeding system, the burner, the vertical furnace and the ash collecting facility. The pulverized fuel K-Tron feeder is used to provide a constant fuel flow rate to the reactor system. A screw feeder at the bottom of the hopper gently moves the bulk material to the large throat and then into the discharge screws. A twin screw system is used since it is particularly useful for handling difficult and hardly flowing fuels. The pulverized fuel is transported pneumatically to the burner (2) in an air stream coming from an air injector.

The furnace (3) consists of a vertical cylindrical combustion chamber that is fired from the top. It is divided in a high-temperature, 2.2 m-long radiative section of 0.3 m diameter made of a refractory material and a low-temperature, 1.8 m-long convective section of 0.25 m diameter which is made of a carbon steel. The radiative section is equipped with four heating elements to compensate for heat losses. The elements allow for setting up a required temperature profile along the radiative section. The heating elements have a thermal input of 8.8 kW each and withstand temperatures up to 1600 °C. The convective 1.8 m-long section is water-cooled using three independent jackets, which allow for co-current, counter-current and cross-flow cooling installations. The convective section's main purpose is to cool down the offgas to temperatures below 1000 °C. A direct quench system (4) is also installed to cool down the offgas to about 180 °C by means of pressurized air. The cooling is necessary to avoid exceeding operating temperatures of downstream equipment. Fly ash is collected in the offgas system consisting of a cyclone (5) and a filter (6) located just downstream the cyclone, to remove fine particles. As all above units cause pressure drops in the offgas system, a vacuum unit (7) is required to drive the flow through all reactor parts.

The properties of the RDF and the two coals fired are given in Table 1. One interesting facet is the relatively low fuel-nitrogen content (0.76%) and low sulfur content (0.04%) of the RDF. The particle size distribution of a South African bituminous Middleburg coal (Fig. 2) corresponds to a typical milling for such coals (75% < 75 μm). A German sub-bituminous Hambach coal is typically milled to coarser fractions (75% < 150 μm), as shown in Fig. 2. For our experiments the RDF has been shredded and milled to provide two size classes: particles smaller than 0.5 mm and not larger than 1 mm, as shown in Fig. 2, to facilitate the feeding through the burner. The RDF fired in our experiments is identical to the one used by Kupka et al. [8] and Morris et al. [9].

The burner is operated at a constant 15 kW fuel input. Fig. 3 shows the burner for the RDF flame (configuration B-1) and for pulverized coal flames (configuration B-2). In configuration B-2,

Table 1
Fuel properties and burner inputs.

Fuel		RDF	Middleburg coal	Hambach coal
Proximate and ultimate analysis, calorific value				
Moisture (as fired)	%	4.15	4.93	9.7
Ash	%	9.34	10.64	4.2
Volatile matter	%	76.59	31.14	45.8
C (as fired)	%	53.29	68.67	60.1
H	%	7.95	4.66	4.7
S	%	0.04	0.83	0.3
N	%	0.76	1.78	0.8
O	%	23.61	8.49	20.3
LCV	MJ/kg	24.29	26.5	22.6
General conditions of investigation				
Air ratio	–	1.2	1.2	1.2
Fuel thermal input	kW	15	15	15
Fuel feed rate	kg/h	2.3	2.1	2.4
Air flow rate	$N\,m^3/h$	16.8	17.4	17.1
Burner velocity (primary air)	m/s	35	30	30
Burner velocity (secondary air)	m/s	7.5	5.8	5.6
Air preheat. temp. (secondary air)	°C	250	250	250
Wall temperature	°C	1200	1200	1200

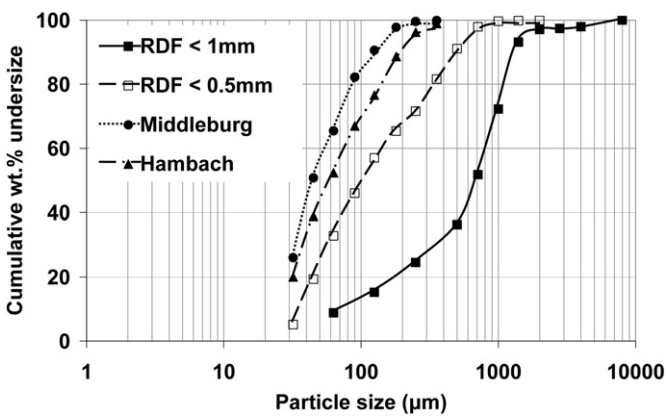

Fig. 2. Size distribution of the fuels fired.

pulverized coals are supplied through an annulus and injected into the reactor with 30 m/s velocity (Table 1). In configuration B-1, the RDF is supplied through the center pipe and is injected with a velocity of 35 m/s. When Middleburg coal is fired using configuration B-1 the injection velocity of 35 m/s is retained. The combustion (secondary) air stream is preheated to a temperature of 250 °C. No swirl is applied: thus, very simple burner configurations are used. The flows into the burner are controlled with a very good accuracy. The fuel feeding rate is maintained within 4% inaccuracy margin while the combustion air flows are controlled within 1% margin. Natural gas shown in Fig. 3 is used for preheating the reactor only and the RDF and coal flames presented in this paper have been obtained without any natural gas support.

Three openings, which are marked in Fig. 1 as Ports 1, 2 and 3, are used for the measurements. At each port at the furnace center the gas temperature is measured using a suction pyrometer and the measurement error is estimated to be 10–15 °C in the 1150–1330 °C temperature range. The O_2, CO, CO_2 and NO concentrations are measured using a gas sampling probe and a set of gas analyzers. The furnace wall temperature is set to 1200 °C and the deviations along the length of the furnace are not larger than ±15 °C. A specially designed sampling probe with a rapid aerodynamic quenching allows for the measurements of carbon burnout. In Fig. 1, at each sampling port, residence times are provided. These have been obtained by performing CFD simulations of trajectories of individual coal particles and by subsequent averaging the obtained residence times over a large number of particles. The flow turbulence was taken into account using the particle tracking procedure developed at the beginning of eighties at the University of Sheffield [16,17] which was later on implemented into the CFD Fluent code. Thus, the residence time figures shown in Fig. 1 correspond to the mean residence time of coal particles and these are indeed indicative.

4. Results and discussion

The results present data for five flames; two RDF flames issued using B-1 burner configuration, two flames of bituminous Middleburg coal issued from B-1 and B-2 configurations and the

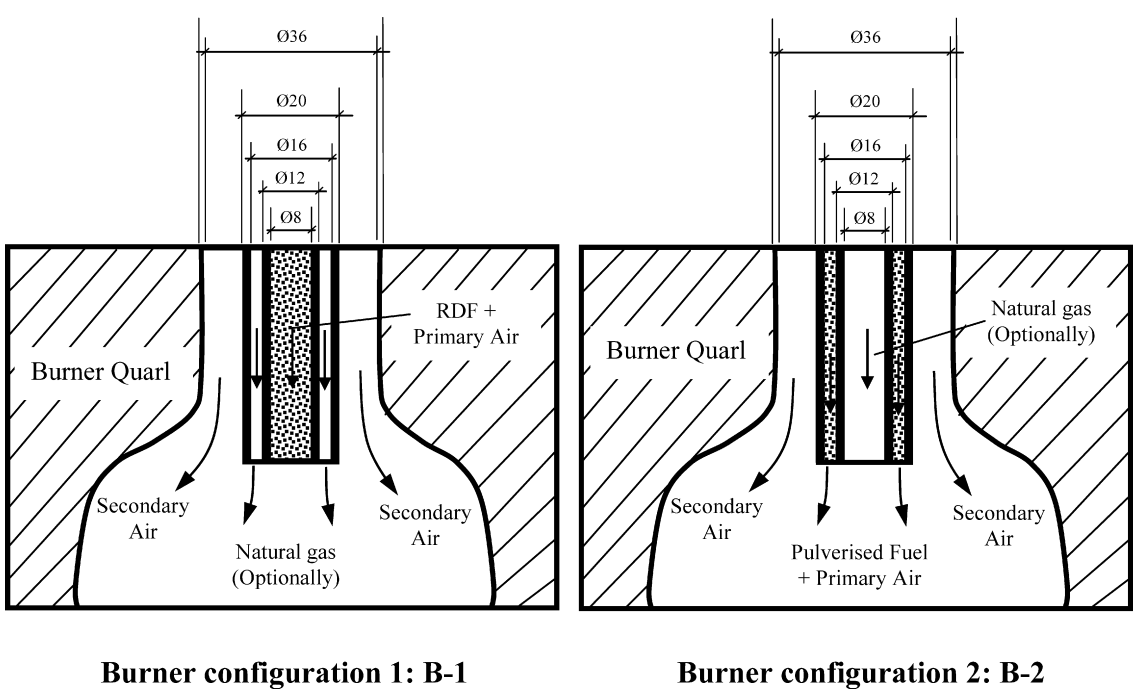

Fig. 3. The burner: Left – configuration for RDF firing; right – configuration for coal firing.

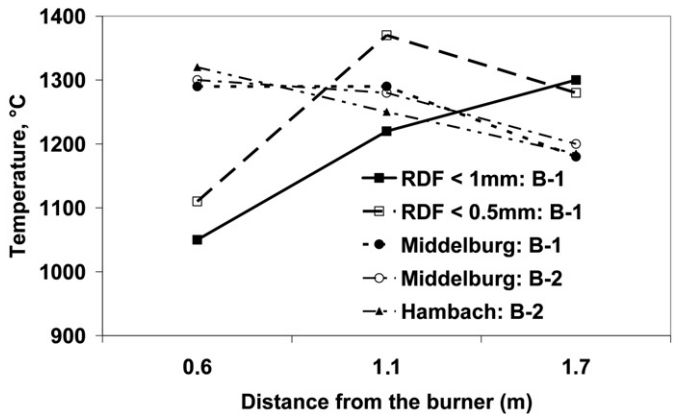

Fig. 4. Temperature profile along the furnace during combustion of the RDF and the two coals.

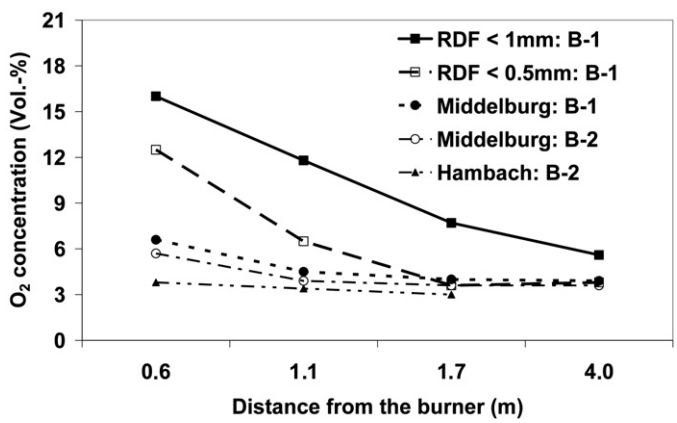

Fig. 5. Oxygen concentration along the furnace during combustion of the RDF and the two coals.

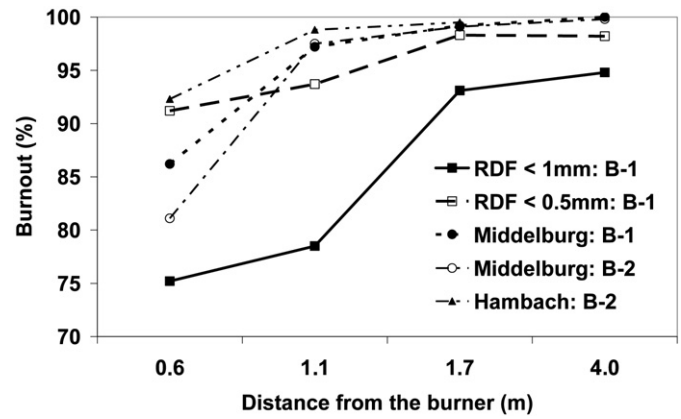

Fig. 6. Char burnout along the furnace.

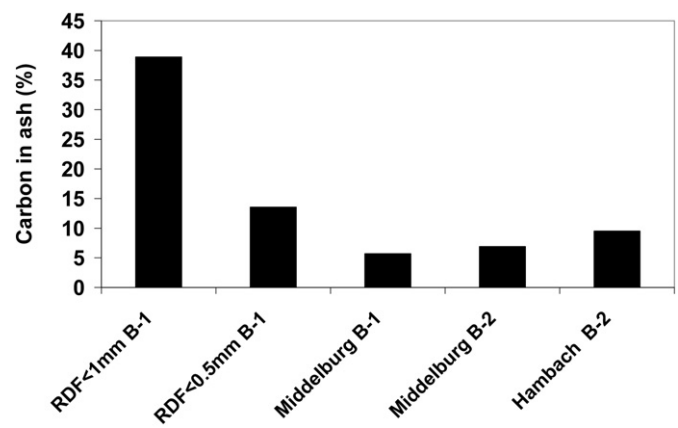

Fig. 7. Carbon in ash in solid matter sampled at Port 3 (1.8 s residence time).

sub-bituminous Hambach coal flame generated using B-2 burner configuration. All the flames have been issued so as to maintain a similar aerodynamics close to the burner and the excess air ratio is kept constant at a value of 1.2. The latter corresponds to 4% oxygen mole fraction (dry) in combustion products. The coal flames serve as references for the RDF flame.

4.1. Flame properties

Figs. 4 and 5 show the measured gas temperature and oxygen concentrations at the centerline of the reactor, respectively. At the first port (300 ms residence time), the temperatures of the coal flames have been measured to be in the range 1300–1330 °C whereas the temperature of RDF flames is around 200–300 °C lower. While the coal flames temperatures gradually decrease along the reactor from 1350 °C to around 1200 °C, the RDF flame temperatures increase from 1050–1100 °C reaching a 1300 °C level far downstream in the radiant section for the flame of coarse RDF. For the flame of fine RDF the maximum flame temperature occurs 1.1 m downstream of the burner. At the first port the oxygen concentrations for the coal flames have been measured to be in the 4–6% range while for the RDF flames as much as 12–16% O_2 has been measured. Thus, Figs. 4 and 5 already indicate fundamental differences between the coal flames and the RDF flames. The pulverized coals burn in the close vicinity of the burner and most of the combustion is almost completed by the first measurement port. To the contrary, the RDF combustion extends far into the reactor and 4 m downstream (1.8 s residence time) 6% O_2 has been measured indicating that the combustion process is still in progress. It is also clear that the main differences in the examined flames originate from the differences in the fuels burned. The influence of burner configuration on the combustion process is visible as one would expect (see for example Clark et al. [18]), however this is a second order influence if compared to the effect of the fuel fired (coals versus RDF), as shown in Figs. 4 and 5.

One of the most important criterions for combustion system design is char burnout. The solid matter collected at the three measurement ports and in the cyclone have been analyzed for their ash content and, assuming that the ash remains inert, the degree of burnout (b) have been calculated as follows

$$b = \frac{1 - \frac{ash_0}{ash_1}}{1 - ash_0},$$

where ash_0 and ash_1 stand for the ash content in the raw fuel sample and in the collected char sample, respectively. As shown in Fig. 6, the RDF burnout at the end of the furnace remains at 93% and 97% for the coarse and fine RDF fractions, respectively, whilst 99.5% figures are applicable to the coal flames. In terms of carbon in ash content which is one of the criteria determining the fly ash utilization, such differences are huge as highlighted in Fig. 7.

The chemical analysis of the RDF (see Table 1) indicates that it is a volatile fuel since the ASTM volatile matter content is as high as 76.59%. Little is known about the process of heating up a 500–1000 μm fluff "particle" suspended in an air flow and exposed to a 200 kW/m^2 heat flux. It has been established [11,12,15] that at 10–20 °C/min particle heating rates, around 80% of RDF volatiles are given off in a 250–450 °C temperature range and the other 20% is released before the particles reach 750 °C. One would expect that under rapid heating conditions prevailing in the reactor, perhaps even more volatiles would be given off. Thus, the RDF

coke yield should not be larger than around 15–20%. The calorific value of RDF volatiles is another important parameter and it has been shown that figures in the range of 16–18 MJ/kg [15] are applicable and these are to be compared with the LCV of raw RDF of 24 MJ/kg. Thus, 70–80% of the RDF chemical enthalpy is released together with volatiles and ignitability of these volatiles together with the burner fluid dynamics (mixing of the volatiles with the air stream) play a key role in determining the flame ignition distance. Regrettably, we are in no position to measure the chemical composition of volatiles in our furnace due to the lack of a reliable measurement technique. However, we have observed a considerable amount of RDF tars which have not been destroyed in 1200–1400 °C flame temperatures but they condensed on cold surfaces (see below). No condensation of tars has been observed in the pulverized coal flames. For the sub-bituminous coal fired, the high temperature volatile matter yield is around 60% and its calorific value has been measured to be around 18 MJ/kg. For the bituminous coal the high temperature volatile matter yield is around 43% and the LCV of volatiles is around 40 MJ/kg [19].

The colliery of the above discussion is the observation that both release and combustion of volatile matter including the tars are of paramount importance in RDF combustion through flames. Little is known about these processes proceeding under high heating rates and the development of appropriate measurement equipment and methods is required. To underline this point further we quote the results of Walasek [20] who measured the chemical composition of volatiles of Middleburg coal and of the RDF used in our experiments. The volatiles were generated under a low heating rate of 4–6 K per minute. For the Middleburg coal, 16.7% (daf) of its original mass was given off as gaseous species (5.1% CO, 4.2% CH_4, 3.8% CO_2, 1.3% N_2, 1.3% H_2, 0.1% C_2H_4, 0.6% C_2H_6, 0.2% C_2H_8, 0.12% O_2) while 15% as species condensable at a temperature of 6 K of which 7.2% was pyrolysis water and 7.8% were tars. For the RDF, 21.4% (daf) of its original mass was given off as gaseous species (4.15% CO, 3.77% CH_4, 8.54% CO_2, 1.08% N_2, 0.76% H_2, 1.05% C_2H_4, 1.29% C_2H_6, 0.5% C_2H_8, 0.24% O_2) while 46.4% as condensable species of which 13.6% was pyrolysis water and 32.8% were tars.

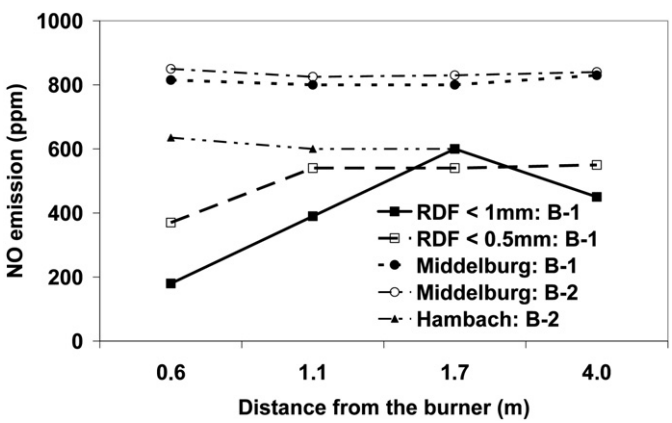

Fig. 8. Measured NO concentrations.

4.2. NO and CO emissions

The measured NO concentrations along the furnace, Fig. 8, substantiate again the already underlined differences between the coal flames and the RDF flame. In the coal flames the NO is formed within the flame zone and 600–700 ppm values have been measured at Port 1. As one would expect, these values remain approximately constant along the furnace since no technique/mechanism is available for its reduction. The NO emissions of the RDF flame keep increasing from around 200 to 500–600 ppm values as the combustion progresses along the reactor. The carbon monoxide concentrations of the coal flames decrease rapidly from around 2000 ppm (vol., dry) at Port 1 to a few ppm at Port 4. However, for the RDF flames the CO levels remain higher than 2000 ppm along the whole length of the 4 m long furnace.

4.3. Ash deposition

A ceramic deposition probe has been inserted in Port 5 located in the convective section of the furnace. Within a relatively short time of 2 h a 15 mm thick layer of a deposit has been accumulated, see Fig. 9, when the coarse RDF fractions were combusted. The deposit containing around 18% of unburned carbon can be classified as a medium hard deposit which may not be so easily removed by

Fig. 9. Deposit accumulated over a period of 2 h on a deposition probe inserted in Port 5.

soot blowing. The rate of growth of such a deposit is even higher than that for a slagging coal. The presence of unburned vapor tars that condense on cold surfaces seems to be a factor enhancing the deposit growth. If such a RDF is co-fired with a non-slagging coal, the RDF presence even in small amounts is likely to promote the deposit formation. Although Kupka et al. [8] have already reported on such a behavior, an understanding of the deposition mechanism is lacking.

5. Further discussion

One should realize that our work is perhaps the first attempt to characterize RDF flames. In this paper we would like to identify issues relevant to engineering and science of such flames since detailed mechanistic explanations cannot be provided. There is no doubt, however that understanding of pure RDF flames is going to be useful in engineering of co-firing systems. As pointed out earlier, one of the most important issues are the volatile matter yield, composition and combustion including the oily tars. The question is at what (minimum) flame temperatures do the RDF tars undergo complete destruction? Perhaps by using the high temperature air combustion technology [21,22] one can boost flame temperatures and increase the residence time in the high temperature region so that a complete tar destruction is achieved?

The issue of RDF burnout is indeed complex, however it is already apparent that the size of the fluffy particles is important. It is unknown how these fluffy particles fly when they are suspended in or advected with an air flow. One may perhaps expect that small RDF particles which we have used in our experiments, fly with an air flow as light spheres. In industrial applications RDF fluff streams contain thin “feather like pieces” of up to one inch size whose aerodynamics is completely unknown. Furthermore, the processes of heating up such pieces of fluff by convection and radiation have to be understood. As indicated in our experiments slagging and fouling issues are of paramount importance.

6. Conclusions

A newly constructed experimental furnace has been used to issue jet flames of a pure refuse derived fuel and to compare their properties with properties of pulverized coal flames. The experiments have demonstrated fundamental differences between the coal flames and the RDF flames. The pulverized coals ignite in the close vicinity of the burner and most of the combustion is completed within the first 300 ms. Despite the high volatile content of the RDF, its combustion extends far into the furnace and after 1.8 s residence time only a 94% burnout has been achieved. This effect has been attributed not only to the larger particle size of fluffy RDF particles but also to differences in RDF volatiles if compared to coal volatiles. Substantial amounts of oily tars have been observed in the RDF flames even though the flame temperatures exceeded 1300 °C. The presence of these tars has drastically enhanced the slagging propensity of RDF flames as rapidly growing deposits of high carbon content have been observed.

Co-firing of RDFs and coals in power station boilers will proceed at temperatures around 300 °C higher than those observed in our experiments and the RDF share is likely to be not larger than 20% by thermal input. The residence time of fuel particles in such a boiler is typically in the range 3–4 s which is twice as large as in our experiments. However, industrial fluff is going to be substantially coarser than the fluff used in our work. All together there are strong indications that even a moderate share of RDF may result in a drop of the boiler efficiency due to increased temperatures downstream of the combustion chamber. Moreover slagging and fouling problems are likely to intensify.

Our paper is perhaps the first attempt to characterize pure RDF flames and there are numerous issues in engineering and science of such flames which require attention. These involve the particle heating up process, devolatilization and combustion of volatiles as well as RDF char burnout. However, understanding of the aerodynamics of fluffy RDF particles in turbulent flows is a prerequisite to successful co-firing.

Acknowledgments

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